



Gamified Tasks Dashboard

Standalone Setup and Demo Validation Runbook

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MAINTAIN DOCUMENT CONTROL - CHECK SHAREPOINT FOR THE LATEST VERSION BEFORE REUSE.

Security notice: Do not paste real database passwords, wallet passwords, OCI tokens, Outlook tokens, Jira tokens, or wallet files into this document, SharePoint comments, Teams/Slack threads, or Git. Use placeholders here and share secrets only through the approved secure team channel.

Hackathon Team

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1. Introduction and Objective

This runbook provides the steps required to run the Gamified Tasks Dashboard locally as a standalone application for hackathon demo validation and team handoff. It is intended to accompany the SharePoint demo package and video, and should be readable by a teammate who has the repository and approved secret-sharing access but no prior local setup.

The objective is to standardize local startup, validation, optional integrations, and troubleshooting so that the team can quickly confirm the app is healthy before a demo.

2. Scope

- In scope: local Windows standalone setup for frontend, backend, Oracle Autonomous Database wallet connectivity, optional OCI GenAI, Outlook Calendar sync through Codex, and Jira sync/enrichment through MCP/Codex.
- Out of scope: production deployment, cloud tenancy administration, manager approval workflows, production incident response, or long-term operations ownership.
- Audience: hackathon team members and reviewers using the SharePoint handoff package.

3. Roles and Responsibilities

No manager approval or formal sign-off is required for this hackathon runbook. Each team member is responsible for using their own local credentials and their own Outlook, Jira, and OCI authentication sessions.

Role	Responsibility
Hackathon team	Maintain this runbook, validate demo startup, and keep the SharePoint handoff current.
Individual teammate	Run local authentication for OCI, Outlook, and Jira when using optional integrations.
Reviewer / evaluator	Use the runbook to start the app locally and validate the smoke-test checklist.

4. Operational Process Prerequisites

4.1 Application components

Area	Required for app startup?	Notes
Frontend	Yes	React production bundle served locally on port 3000.
Backend	Yes	FastAPI service on port 8000.
Oracle ADB wallet	Yes for real data	Required for Oracle-backed tasks, auth, quests, insights, focus, and overviews.
Oracle DB credentials	Yes for real data	Use the DEVQUEST_APP schema. Share the password out-of-band only.
OCI GenAI	Optional	Real AI requires local BOAT OCI CLI session authentication. Otherwise the app starts in mock AI mode.
Outlook Calendar Codex plugin	Optional	Needed only for Outlook calendar sync through Codex.
Jira MCP	Optional	Needed only for Jira sync and Jira RCA/enrichment flows through Codex.

4.2 Local prerequisites checklist

- Repository is cloned locally.
- Windows PowerShell or Command Prompt is available.
- Node.js/npm is installed for frontend build/start.
- Python is installed for backend virtual environment.
- Oracle wallet is available through the approved secure channel when real data is required.
- Oracle Instant Client is installed when Oracle thick mode is used.

5. Operational Process Overview

The preferred path is the one-command startup from the repository root. Manual frontend and backend startup steps are included for troubleshooting or development-mode testing.

Task	Preferred method	When to use
Start full app	<code>.\start-devquest.cmd</code>	Normal demo validation and local handoff.
Stop full app	<code>.\stop-devquest.cmd</code>	After demo validation or before restarting due to port conflicts.
Start frontend only	<code>npm start</code>	Frontend development server testing.
Start backend only	<code>uvicorn main:app</code>	Backend debugging or API validation.
Validate health	Browser and Invoke-WebRequest checks	After every startup.

6. Standard Startup and Shutdown

6.1 Recommended one-command startup

Run from the repository root:

```
.\start-devquest.cmd
```

The launcher performs the following startup tasks:

- Verifies backend and frontend folders.
- Creates the backend virtual environment if missing.
- Checks Oracle wallet and Oracle Instant Client paths.
- Sets Oracle DB environment variables.
- Detects whether a BOAT OCI CLI security-token session exists.
- Uses real OCI GenAI when BOAT session auth exists; otherwise uses mock AI.
- Builds the frontend production bundle when needed.
- Starts FastAPI on `http://127.0.0.1:8000`.
- Serves the production frontend on `http://127.0.0.1:3000`.

6.2 Stop the app

```
.\stop-devquest.cmd
```

7. Health Checks and URLs

Purpose	URL	Expected Result
Gamified Tasks Dashboard app	<code>http://127.0.0.1:3000</code>	Frontend loads.
Backend root health	<code>http://127.0.0.1:8000/</code>	<code>{"msg": "DevQuest Pro"}</code>
Swagger UI	<code>http://127.0.0.1:8000/docs</code>	Interactive API docs load.
Swagger shortcut	<code>http://127.0.0.1:8000/swagger</code>	Swagger docs load or redirect.
ReDoc	<code>http://127.0.0.1:8000/redoc</code>	ReDoc API reference loads.
OpenAPI JSON	<code>http://127.0.0.1:8000/openapi.json</code>	OpenAPI JSON returns successfully.

8. Frontend Standalone Procedure

Use this procedure only when running the React development server instead of the production static server.

```
cd frontend
npm install
npm start
```

Frontend development URL: `http://localhost:3000`

Production build only:

```
.\build-devquest-prod.cmd
```

or

```
cd frontend
npm run build
```

The production bundle is created under frontend/build.

9. Backend Standalone Procedure

Use this procedure only when starting the backend manually instead of using start-devquest.cmd.

```
cd backend
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install --upgrade pip
.\.venv\Scripts\python.exe -m pip install -r requirements.txt
.\.venv\Scripts\python.exe -m uvicorn main:app --host 127.0.0.1 --port 8000
```

Backend URL: http://127.0.0.1:8000

10. Oracle Database Connectivity

10.1 Oracle wallet setup

Oracle Autonomous Database uses wallet-based connectivity. Store wallet files outside the repository.

Recommended local wallet folder:

```
%USERPROFILE%\oracle\wallet_tasksdb
```

Expected files include tnsnames.ora, sqlnet.ora, and ewallet.pem.

```
New-Item -ItemType Directory -Force -Path "$env:USERPROFILE\oracle\wallet_tasksdb"
Expand-Archive -LiteralPath "$env:USERPROFILE\Downloads\Wallet_TASKSDB.zip" -DestinationPath
"$env:USERPROFILE\oracle\wallet_tasksdb" -Force
```

Use the TNS alias: tasksdb_tp

10.2 Oracle Instant Client setup

The Windows startup script looks for Oracle Instant Client in either of these locations:

```
C:\oracle\instantclient_23_0
%USERPROFILE%\Downloads\instantclient-basic-windows.x64-23.26.1.0.0\instantclient_23_0
```

The folder must contain oci.dll. The launcher starts the backend in Oracle thick mode using ORACLE_DB_THICK_MODE and ORACLE_CLIENT_LIB_DIR.

10.3 Oracle DB runtime variables

Use the runtime schema DEVQUEST_APP. Do not use the ADMIN schema for local app runs unless the team explicitly asks for it.

```
$env:DEVQUEST_DATA_MODE="oracle"
$env:DB_USER="DEVQUEST_APP"
$env:DB_PASSWORD="<shared out-of-band DB password>"
$env:DB_DSN="tasksdb_tp"
$env:DB_WALLET_DIR="$env:USERPROFILE\oracle\wallet_tasksdb"
$env:DB_WALLET_PASSWORD="<shared out-of-band wallet password>"
$env:TNS_ADMIN="$env:USERPROFILE\oracle\wallet_tasksdb"
$env:ORACLE_DB_THICK_MODE="1"
$env:ORACLE_CLIENT_LIB_DIR="<instant_client_folder>"
$env:DB_POOL_SIZE="10"
$env:DB_POOL_MIN="1"
$env:DB_POOL_MAX="10"
$env:DB_POOL_INCREMENT="1"
```

11. OCI GenAI Configuration

Real AI is optional for local startup. The launcher uses the following behavior:

Local condition	AI mode used
BOAT OCI CLI profile exists and has a security_token_file	real
BOAT profile/token is missing	mock
DEVQUEST_AI_MODE is explicitly set before startup	Uses the explicit value

Authenticate with OCI CLI for real AI:

```
oci session authenticate --region us-phoenix-1 --tenancy-name bmc_operator_access --profile-name boat
```

Expected generated profile:

```
[boat]
fingerprint=<generated by OCI CLI session auth>
key_file=<path to .oci/sessions/boat/oci_api_key.pem>
tenancy=<bmc_operator_access tenancy OCID>
region=us-phoenix-1
security_token_file=<path to .oci/sessions/boat/token>
```

Real AI variables:

```
$env:DEVQUEST_AI_MODE="real"
$env:DEVQUEST_AI_PROVIDER="oci_genai"
$env:OCI_AUTH_TYPE="security_token"
$env:OCI_CONFIG_PROFILE="boat"
$env:OCI_REGION="us-phoenix-1"
$env:OCI_COMPARTMENT_ID="<team_aurora_compartment_ocid>"
$env:OCI_GENAI_ENDPOINT="https://inference.generativeai.us-phoenix-1.oci.oraclecloud.com"
$env:OCI_GENAI_SERVING_MODE="on_demand"
$env:OCI_GENAI_REQUEST_FORMAT="generic"
$env:OCI_GENAI_MODEL_ID="google.gemini-2.5-flash"
```

12. Codex Integrations

12.1 Outlook Calendar plugin in Codex

Outlook Calendar sync uses the Codex Outlook Calendar plugin, not a backend-stored Microsoft app secret.

Item	Value
Source	Outlook Calendar
Backend sync endpoint	/api/v1/sync/runs
Codex capability expected	Outlook Calendar list_events

1. Install or enable the Outlook Calendar plugin in Codex.
2. Complete Outlook sign-in through the Codex plugin flow.
3. Start Gamified Tasks Dashboard normally.
4. Use the Sync page/action in the app to fetch Outlook Calendar events.

12.2 Jira MCP setup

Jira sync and Jira RCA/enrichment use Codex plus the Jira/Confluence MCP server.

Setting	Value
Expected MCP server name	central_jira_confluence
Runtime setting	\$env:JIRA_MCP_SERVER="central_jira_confluence"

5. Install or configure the central_jira_confluence MCP server in Codex.
6. Authenticate to Jira through the MCP flow.
7. Keep Jira credentials and tokens outside the repository.
8. Start Gamified Tasks Dashboard normally.
9. Use the Sync page/action or Jira AI enrichment flow from the app.

If Jira authentication is missing, the app may report that Jira MCP authentication is required. Complete Jira SSO, then retry the sync or enrichment action.

13. Smoke Test Checklist

13.1 API checks

```
Invoke-WebRequest -Uri "http://127.0.0.1:8000/" -UseBasicParsing
Invoke-WebRequest -Uri "http://127.0.0.1:8000/openapi.json" -UseBasicParsing
```

13.2 Browser checks

- Open <http://127.0.0.1:3000>.
- Log in or register with the test user flow.
- Confirm Dashboard loads.
- Confirm My Tasks loads.
- Open AI Insights.
- Click AI.
- If real OCI is configured, verify a generated insight appears.
- If real OCI is not configured, verify mock AI still lets the app work.
- Open Swagger at <http://127.0.0.1:8000/docs>.
- Open ReDoc at <http://127.0.0.1:8000/redoc>.

14. Common Troubleshooting

Symptom	Likely Cause	Fix
Backend starts but DB endpoints fail	Oracle wallet, Instant Client, DB password, or TNS alias issue	Recheck wallet folder, oci.dll, DB_DSN, and DB credentials.
ORA-01017	Wrong DB username/password	Use DEVQUEST_APP and the shared app-user password.
ORA-12154	TNS alias not found	Confirm tasksdb_tp exists in tnsnames.ora and TNS_ADMIN points to the wallet folder.
AI generate returns auth failure	OCI session expired or missing	Rerun OCI session authentication with profile name boat.
AI uses mock output	BOAT token not found, or DEVQUEST_AI_MODE=mock was set	Authenticate with OCI CLI or explicitly set DEVQUEST_AI_MODE=real.
Outlook sync fails	Codex Outlook Calendar plugin not signed in	Sign in through Codex plugin flow and retry.
Jira sync/enrichment fails	Jira MCP not configured or SSO not complete	Configure central_jira_confluence, complete Jira auth, and retry.
Port already in use	Existing backend/frontend still running	Run .\stop-devquest.cmd, then .\start-devquest.cmd.

15. Files to Reference

File	Why it matters
README.md	High-level overview and quick start.
start-devquest.ps1	Source of truth for local integrated startup.
start-devquest.cmd	Windows launcher wrapper.
build-devquest-prod.ps1	Production frontend build helper.
docs/local-start.md	Local start details, API docs, DB and AI environment examples.
docs/oracle-db-wallet-setup.md	Oracle wallet and connection pool setup.
docs/oci-genai-team-setup.md	BOAT OCI CLI and GenAI setup.
backend/config.py	Runtime environment variable readers.
backend/db.py	Oracle pool setup and connection helper.
backend/services/sync_service.py	Jira and Outlook sync behavior.
backend/codex-config.py	Jira MCP and Codex runtime settings.

16. Definitions

Term	Definition
ADB	Autonomous Database

BOAT	OCI CLI session profile used for local GenAI authentication
Codex	Local integration surface used for Outlook Calendar and Jira/MCP actions
DSN	Database service name used by Oracle client connectivity
GenAI	Generative AI service used for real AI insights when configured
MCP	Model Context Protocol server used for Jira/Confluence access
TNS_ADMIN	Oracle environment variable that points to the wallet directory

17. Revision History

Author	Version	Comment	Date
Hackathon team	1.0	Formalized runbook using Oracle-style title page, linked contents, standalone setup flow, validation checklist, and troubleshooting.	2026-05-08